

A Genetic Algorithm (GA) is used to maximize the fitness of binary strings of length 6.

Each chromosome represents a candidate solution.

The fitness of each chromosome is defined as:

$f(x)$ =decimal value of the binary chromosome

Chromosome Binary String

P1	101101
P2	111000
P3	010111
P4	110101
P5	100110

Use the random numbers generated sequentially where needed:

0.12, 0.41, 0.67, 0.88, 0.63, 0.27, 0.81, 0.33, 0.58, 0.14

(a) Calculate the fitness scores

(b) Compute the selection probabilities and determine two pairs of parents using (proportionate) roulette wheel selection.

(c) Perform single point crossover of the selected parents to create the offspring. (position= $\lceil r \times 6 \rceil$)

(d) Apply one-bit mutation on each of the produced offspring. (position= $\lceil r \times 6 \rceil$)

(e) Compute the fitness value of each offspring

(f) Given the value of Elitism = 0.4 and Generation Gap = 0.6, find the chromosomes that will make the population for the next generation($t+1$).

Solution:

(a)

Chromosome	Binary	Fitness
P1	101101	$32 + 8 + 4 + 1 = 45$
P2	111000	$32 + 16 + 8 = 56$
P3	010111	$16 + 4 + 2 + 1 = 23$
P4	110101	$32 + 16 + 4 + 1 = 53$
P5	100110	$32 + 4 + 2 = 38$

(b)

Total Fitness

$T = 45 + 56 + 23 + 53 + 38 = 215$

Chromosome	Fitness	Probability	Cumulative
P1	45	0.2093	0.2093
P2	56	0.2605	0.4698
P3	23	0.1070	0.5768

Chromosome	Fitness	Probability	Cumulative
P4	53	0.2465	0.8233
P5	38	0.1767	1.0000

(parents are selected based on the random numbers lying in cumulative ranges)

Random numbers: **0.12, 0.41, 0.67, 0.88**

Selection:

- 0.12 → **P1**
- 0.41 → **P2**
- 0.67 → **P4**
- 0.88 → **P5**

Parents selected:

- Pair 1: (**P1, P2**)
- Pair 2: (**P4, P5**)

(c)

(Since there was a bit confusion with the selected bit, please follow the lecture slides where the crossover point is after the position selected. E.g if position=2 -> crossover after position 2 i.e between bit 2 and 3)

Position: [r×6]

Pair 1 (r = 0.63)

$[0.63 \times 6] = [3.78] = 4$

P1 = 1011|01

P2 = 1110|00

Offspring:

- O1 = 101100
- O2 = 111001

Pair 2 (r = 0.27)

$[0.27 \times 6] = [1.62] = 2$

P4 = 11|0101

P5 = 10|0110

Offspring:

- O3 = 110110
- O4 = 100101

(d)

(here the position obtained by the formula should be mutated)

Offspring	r	Position	Result
O1 = 101100	0.81	5	101110
O2 = 111001	0.33	2	101001

Offspring	r	Position	Result
O3 = 110110	0.58	4	110010
O4 = 100101	0.14	1	000101

(e)

Offspring	Binary	Fitness
O1	101110	$32+8+4+2 = 46$
O2	101001	$32+2+1 = 41$
O3	110010	$32+16+2 = 50$
O4	000101	$4+1 = 5$

(f)

- Population size = 5
- Elitism = $0.4 \times 5 \rightarrow 2$ best parents carried forward
- Generation gap = $0.6 \times 5 \rightarrow 3$ offspring selected

Step 1: Elites from current population

Top 2 fitness:

- P2 = 56
- P4 = 53

Step 2: Best 3 offspring

Offspring ranked:

- O3 = 50
- O1 = 46
- O2 = 41

Final population (Generation t+1)

{P2, P4, O3, O1, O2}